# An Efficient Burst Traffic-Aware Clustering and Mobility Management model for mobile-sink based Heterogeneous Wireless Sensor Networks

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## Abstract

Wireless sensor networks (WSNs) consist of small, low-cost, low-power sensors deployed to monitor environmental changes and transmit data to a static base station. A major challenge in WSNs is energy consumption, particularly in nodes near the static sink, leading to the "hotspot" or "energy hole" problem, which disrupts data transmission and reduces network lifespan. Traditional approaches with static sinks and multi-hop routing often result in premature node failure and data loss. This paper introduces an Efficient Burst Traffic-Aware Clustering and Mobility Management (EBTCM) model to optimize the energy efficiency and data collection process in mobile sink-based heterogeneous WSNs (mWSNs). The proposed model is particularly effective in managing burst traffic, where sudden surges of data must be collected promptly to avoid loss, and uses Mobile Sinks (MS) to dynamically adjust routes. It leverages clustering techniques and mobile sink routing based on the Travelling Salesman Problem (TSP) to extend the network lifetime, reduce energy consumption, and efficiently manage both regular and burst traffic. Simulation results demonstrate significant improvements in path length reduction, energy efficiency, and network longevity compared to existing approaches, highlighting the potential of the proposed model in real-world WSN applications.

## Keywords

Wireless Sensor Networks (WSNs), Cluster Head (CH), Mobile Sink (MS), Travelling Salesman Problem (TSP), Burst Traffic, Energy Efficiency, Mobility Management.

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## 1 Introduction

Wireless sensors are tiny electronic devices that track environmental changes in real time[12]. They are often low-cost, low-power devices with data processing and transfer capabilities to a Base Station ($B_s$). Following a successful deployment, the sensor nodes self-organize to form a network known as wireless sensor network (WSN)[1]. Typically, replacing battery-powered nodes after deployment is a laborious procedure. Various sectors, automated homes, contemporary healthcare, military deployments, ecological monitoring, and security are just a few of the numerous domains in which sensor technology has been used.

In sensor networks, a $B_s$ is situated at a specific coordinate of sensor network with infinite energy source. Generally, it's an internet-connected computer device. Conversely, mobile sensor devices such as Mobile Sink (MS) is used in mobile WSNs to collect data efficiently, without bothering all neighborhood nodes, which causes energy hole problem while using a static sink. Mobile Sink Node can dynamically relocate itself within a sensing area. Here, MS based wireless sensor networks (mWSNs)represent a significant evolution in wireless sensor network(WSN) technology[1] [5]. Data from risky areas can be collected using MS. mWSNs offer several advantages over traditional static WSNs, making them well-suited for various applications: enhanced coverage and flexibility, improved fault tolerance and resilience, dynamic network reconfiguration, and improved network lifetime. When it comes to environmental monitoring, mWSNs are essential for obtaining data in real time on temperature, humidity, pollution levels, and air quality which offer important insights into environmental phenomena and support disaster management, ecological conservation, and urban planning initiatives by deploying sensor nodes in hazardous or remote environments.

WSNs face significant challenges, such as limited battery life and the need for efficient energy management, alongside difficulties in ensuring reliable communication due to interference and signal issues. mWSNs introduce additional complexities, including need for optimized Data Collection Points (DCPs) selection, which requires energy-intensive calculations. Moreover, determining trajectory and routing for regular data collection from DCPs poses a challenge, as does managing burst traffic, where emergent data must be quickly collected to prevent loss. As the network scales, designing adaptable architectures and efficient protocols becomes increasingly challenging, necessitating effective resource management.

To determine optimal path in a sensor network, two approaches are used: Rendezvous Point (RP) based approach, where mobile sinks (MSs) collect data at designated virtual points, and cluster-based approach[7][11]. The RP-based approach involves significant calculation overhead, hence the cluster-based method more efficient for routing. In cluster-based approach, nodes are grouped into clusters with designated Cluster Heads (CHs) that aggregate data from member nodes and forward it to the $B_s$ via MS. This clustering enhances energy efficiency by balancing the network load and reducing congestion through multiple routing options.

After clusters and CHs are formed DCPs are initialized and the route is formed for first round. The MS traverses from one DCP to another based on minimum distance. This process is implemented using the renowned Travelling Salesman Problem (TSP)[8] algorithm which uses greedy approach to visit point to point. We call this regular data collection, and this process continues until there is burst data generated at place(s) in the network[13]. In mWSNs, when significant data bursts occur, the MS must quickly adjust its trajectory to prevent data loss, a process known as trajectory changing. This work focuses on developing an energy-efficient clustering method with minimal calculation overhead and a routing algorithm that optimizes path traversal for both regular and burst data collection, ultimately enhancing the network's energy efficiency and mobility management.

Major contributions are summarized as follows:

(1) **Clustering and Cluster Head Selection:** We propose clustering, and, CH selection method based on some criteria such as- maximum residual energy, maximum number of neighbor nodes, and minimum distance from the $B_s$ to enhance network scalability, energy efficiency, and data collection reliability in WSNs.
(2) **Selection of Data Collection Points:** After formation of CHs in the network, data collection point of the network is selected.
(3) **Routing on Burst Data Generation:** The process of trajectory changing for collecting burst data as soon as it's generated in network is introduced in this work.
(4) **Simulation of Network:** We simulated the entire sensor network and routing of the mobile sink in Matlab. The simulation shows our proposed routing strategy is more efficient in terms of data collection, routing, and sink mobility management compared to other algorithms in literature.

Overall, our contributions aim to address the existing challenges in WSNs and pave the way for more efficient and reliable data collection, routing, and sink mobility management in these networks. Rest of the paper is formulated as follows: Chapter 2 interprets Literature Review. Chapter 3 analyzes the Proposed Methodology. Chapter 4 shows Performance Evaluation. Chapter 5 suggests some future research directions and the paper is concluded in this work.

## 2 Literature Review

An energy-saving data collection algorithm is proposed by Zhang et al. [14], which uses mobile sinks in WSN to achieve extended network lifetime by balancing energy consumption between the sensor nodes. Their proposed method builds the structure of the clusters with member nodes and routes the calculated data to assigned rendezvous points. Another routing algorithm that assembles several trajectories to reduce energy consumption and latency caused by an MS. is proposed by Alsaafin et al. [3]. The algorithm is employed in four major stages: data detection, rendezvous point selection[11], design of the path, and data transmission. As they used only one MS exclusively, the data collection is not prioritized, and packet loss occurs due to buffer overflow in the network. The mobile sink's energy efficiency is not taken into account in this work.

Ling and Cheng [15] introduced dynamic path planning for mobile sink nodes, where when burst data[13] occurs, waking up neighboring nodes to cache data is introduced. In each round of collecting data by the mobile sink, re-electing cluster heads is also introduced to collect data from the calculated points efficiently. The observation here is, they used multi-hop for transferring data, which uses more energy, and only energy level is considered for selecting CHs, which is not an efficient way in the long run.

Al-Kaseem et. al. [2] presented a way of partitioning the network where the sensing field is partitioned into equal regions depending on the number of sinks to avoid the energy-hole problem. They also introduced the Stable Election Algorithm (SEA), which minimizes the message exchange between sensor nodes. The observation of their work is that the network region can be a sparse region where equal partition may not be necessary.

An energy-efficient and dependable routing system for data transmission in a heterogeneous sensor network was presented by Maurya et al. [10]. To guarantee prompt data delivery throughout the network, a restricted search space is established in the suggested method. Additionally, a protocol that chooses a suitable method for quick data transmission has been established. However, this approach is not designed for big networks that consist of several MSs.

A data dissemination algorithm is introduced by Lee et al. [9] taking into consideration the routing and shape of movement of a Mobile Sink. In this protocol, a routing strategy is devised for paths of MSs as three categories, including regular, directional, and random motions. Although the study's simulation results show that the mobility methods are more effective than earlier research based on energy depletion and package delivery rate, it has been noted that random mobility is the source of excessive delays and path lengths.

In literature most of the works in WSN, creating clusters and choosing cluster heads are based on residual energy of the sensor nodes. Most of them employed single tier energy-driven battery-powered nodes which means all of the sensor nodes have the same initial energy sources when deploying the network. Some of the works use coverage radius of the sensor nodes but they couldn't use efficient use of it. In this work we introduced efficient clustering approach which uses better use of coverage area when selecting the cluster heads. As a result the clustering and proposed routing strategy shows comparatively better results in terms of network lifetime, path length visit of mobile sink, and energy consumption of sensor nodes in the network.

# 3 Proposed Methodology

## 3.1 Network Model and Assumptions

We assume a WSN, consists of a base station or sink $B_s$, a set of sensor nodes $\mathbb{N}$, where a sensor is defined by $n_i \in \mathbb{N}$. Energy of each sensor node is $E_i$. The network is called heterogeneous as there are three different types of nodes. The set of neighbor nodes of a particular node $n_i \in \mathbb{N}$ is $\mathbb{Y}_i$.
Distance of a node $n_i \in \mathbb{N}$ from the $B_s$ is denoted by $dist(n_i, B_s)$, and $\alpha$ is the energy-threshold value to exceed for general nodes, $\beta$ is denoted for the coverage-threshold value to exceed for general nodes.
The heterogeneous nodes are three types- super nodes, advanced nodes and general nodes. Coverage radii of heterogeneous nodes $n_i$, and $n_j$ are denoted by $R_i$, and $R_j$ respectively
Total number of nodes is $|\mathbb{N}|$, where,

super nodes are defined by,

$$\mathbb{N}_S \subseteq \mathbb{N}, where \Big\{ n_i \in \mathbb{N} \land E_i \geq 3\alpha \land R_i \geq 3\beta \tag{1}$$

advance nodes are defined by,

$$\mathbb{N}_A \subseteq \mathbb{N}, where \Big\{ n_i \in \mathbb{N} \land E_i \geq 2\alpha \land R_i \geq 2\beta \tag{2}$$

general nodes are defined by,

$$\mathbb{N}_G \subseteq \mathbb{N}, where \Big\{ n_i \in \mathbb{N} \land E_i \geq \alpha \land R_i \geq \beta \tag{3}$$

The set of all types of nodes is: $\mathbb{N} = \mathbb{N}_S \cup \mathbb{N}_A \cup \mathbb{N}_G$.
For any node $n_i \in \mathbb{N}$,

$$\mathbb{Y}_i = \{n_j\};\ dist(i, j) \leq (R_i + R_j) \tag{4}$$

Here $\mathbb{Y}_i$ is the set of neighborhood sensor nodes for which the distance between the current sensor node $n_i$ and the calculating sensor node $n_j$ is less than or equal to the coverage radius of $n_i$ and $n_j$ sensor nodes. The number of neighbor nodes for a specific sensor node is $|\mathbb{Y}_i|$.

The table 1 contains all of the symbols we used in this work. The symbols are mentioned throughout this document. This table will help finding the symbols easily.

| Parameter | Symbol |
|---|---|
| Sensor nodes | $\mathbb{N}$ |
| Base Station | $B_s$ |
| Mobile Sink | $MS$ |
| Cluster Head | $CH$ |
| Super Node | $\mathbb{N}_S$ |
| Advance Node | $\mathbb{N}_A$ |
| General Node | $\mathbb{N}_G$ |
| Energy threshold | $\alpha$ |
| Coverage radius threshold | $\beta$ |
| Energy of $n_i$ node | $E_i$ |
| Coverage Radius of $n_i$ node | $\mathbb{R}_i$ |
| Number of neighbor nodes | $\lvert\mathbb{Y}_i\rvert$ |
| Cluster Head Eligibility | $\Phi_i$ |

**Table 1: Table of Parameters' Symbols**

In mWSN[1], it's challenging to select the Data Collection Points(DCPs) for a Mobile Sink (MS). For this, we develop clustering method for the MS to collect data from cluster heads[7]. Here, the deployment of mobile sinks can increase the network coverage time and energy usage during periods of burst traffic. Hence, it is crucial to handle this burst traffic using mobile sinks in an energy-efficient mobility manner and our objective is to focus on this area.

## 3.2 Cluster Formation

We assume a network where all the sensor nodes are deployed randomly. Each sensor node has attributes - coverage area, remaining energy, and Cartesian coordinates.

**Cluster Head Selection:** The CH will be selected based on these criteria:

- Energy of individual sensor node exceeds energy threshold value
- Maximum number of neighbor nodes
- Minimum distance from the Base Station, $B_s$

For every sensor node calculate a value $\Phi_i$, where

$$\Phi_i = w_1.\frac{E_i^{Res}}{\alpha} + w_2.|\mathbb{Y}_i| - w_3.dist(n_i, B_s) \tag{5}$$

Here, $\Phi_i$ represents the CH-eligibility value for each node, where $E_i^{Res}$ is the residual energy, and $\alpha$ is the energy-threshold value of node $n_i$. The ratio of $E_i^{Res}$ to $\alpha$ gives a normalized value. $|\mathbb{Y}_i|$ is the number of sensor nodes covered by $n_i$, and $dist(n_i, B_s)$ is the distance from $n_i$ to the base station, which is subtracted from $\Phi_i$. Here, $w_1, w_2, w_3$ are weighing factors.

After calculating $\Phi_i$ for each node, the node with the highest $\Phi_i$ is selected as the CH and will advertise its status to neighboring nodes. These neighbors then evaluate their own $\Phi_i$ values and decide whether to join the existing cluster or form a new one. The locations of the CHs are stored in the set $\mathbb{U}$, which creates DCPs.

Clusters and CHs are formed in each round. At the very beginning all the sensor nodes are given energy based on their types (whether it's a general, or an advance or a super node). Starting from the $B_s$ the MS traverses through the CHs and ends its round by visiting the $B_s$ again. After that, based on the previously remaining energy and other attribute values new clusters and CHs are selected for the next round. This procedure continues until the energy of the network is finished.

### 3.3 Mobile Sink Trajectory Path Selection

After formation of the clusters and CHs, the information will be sent to the $B_s$ from where MS starts travelling. Before starting its journey the MS calculates the distances of the CHs from the current point and take decision.

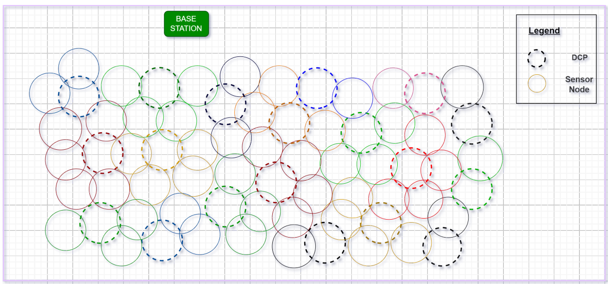


**Figure 1: Data Collection Points Selection**

*3.3.1 Data Collection Points Formation.* In our approach, the CHs are the the Data Collection Points (DCPs). The MS starts from the $B_s$ and moves through the DCPs. The MS first selects the closest DCP to the $B_s$ from a set of unvisited CHs ($\mathbb{U}$) as the first DCP, and then arrives at that point. After collecting data, the DCP is added to the set of visited DCPs ($\mathbb{V}$), and the MS then selects the next closest DCP from $\mathbb{U}$ using a greedy approach based on the Travelling Salesman Problem (TSP). This process continues until the MS returns to the Bs, completing a round. In next round, new DCPs are formed. The process is illustrated in figure 1. However, the scenario changes, when bursty traffic happens. Therefore, in our proposed sstem MS will collect two types of data, i. regular data, ii. bursty data. As a result the traversing path need to be determined accordingly, which are described as below:

**1. Path determination for regular data:**

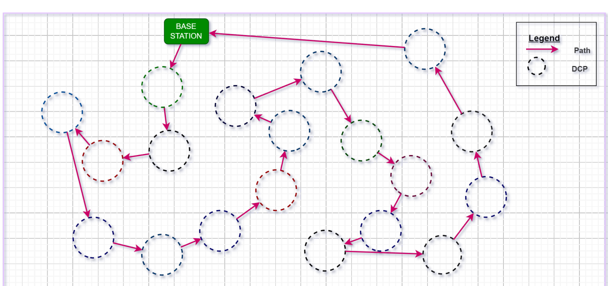


**Figure 2: Route Selection based on Minimum Distance**

The route of the mobile sink needs to be calculated in an efficient manner. The movement of the mobile sink needs energy. To decrease the energy depletion due to the traversing of the mobile sink, the overall path per round needs to be as short as possible. The less the length of the routing path, the less energy consumption will occur. For this, the routing of the mobile sink is a crucial thing to consider for energy-efficiency in the network.

The traversal of a Mobile Sink (MS) will be predefined, with shortest path ($l_s$) consideration. We used modified Travelling Salesman Problem (TSP)[8] for path traversal of a MS. Two types of traversal is defined, regular data collection and burst data collection[13]. Soon after the deployment of a network, the mobile sink(s) will start to visit the nodes (CHs in this case) in regular data collection which is predefined. The trajectory of a MS will suddenly be updated only when there's burst data producing in a region. The burst data collection will be defined in some phases, whether the CH producing burst data is already visited in this round or not, if this happens then how the trajectory will be changed. Again if the CH producing vast data is not visited then how the data will be collected and how the trajectory will be changed. The route selection and traversing is shown in figure 2.

**2. Path determination for bursty data:** In the scenario where burst data occurs at a Data Collection Point (DCP), denoted by $R_b$, its coordinates are stored in variable A. The next DCP to be visited by the mobile sink (MS) during regular data collection is $R_s^n$, with coordinates stored in B. The previous DCP before the burst data point is $R_b^p$ (coordinates stored in C), and the immediate next DCP after burst data is $R_b^n$ (coordinates stored in D). Upon detecting burst data, the MS immediately alters its trajectory to the burst data location, collects the data, and then resumes its path toward the next relevant DCP for regular data collection. The burst data generation scenario shown in figure 3. For example, there may happen some different cases:

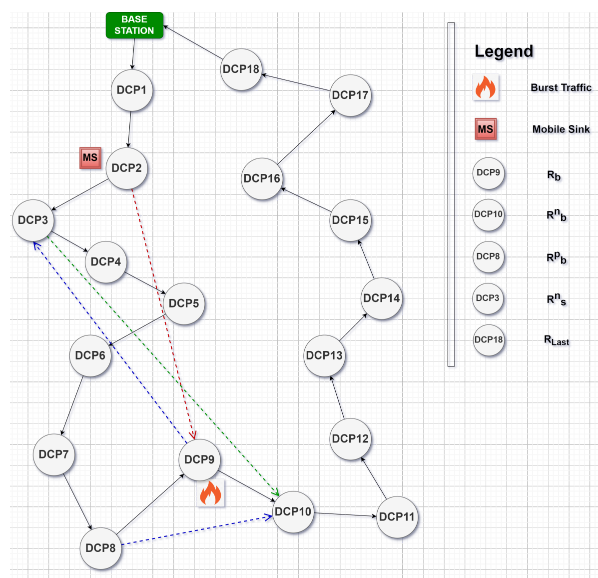


**Figure 3: Trajectory Changing on Burst Data**

**Case 1:** Burst traffic is at the first or last DCPs which are covered by the $B_s$. In this case the $B_s$ will collect the data from the DCPs which are covered by the CH of the DCP and the $B_s$ (overlapped). In the mean time the MS will continue the regular data collection. The cost to traverse the path is 0.

**Case 2:** The MS will go to $R_s^n$, and collect the data in regular data collection method. Then it will traverse through the regular data collection path and stops just before the location (DCP) where burst data occurred which is $R_b^p$. After that it will go to $R_b^n$ without being collected data from $R_b$ as the burst data is already collected and an array is tracking which DCPs are already visited in every round, and then collect data from the rest of the DCPs in regular data collection method(figure 3). The cost to traverse this path is,

$$cost = dist(R_b, R_s^n) + dist(R_b^p, R_b^n) \tag{6}$$

**Case 3:** The burst traffic is at the next DCP: The MS will collect the data immediately just after collecting the data from the current DCP. The cost to traversal for the burst data is 0.

**Case 4:** The MS will collect data from the burst area and then traverse through a predefined regular data collection area and change its trajectory just before visiting the $B_s$, and go directly to the $R_s^n$), and after collecting the data from this DCP it will traverse in predefined order through $R_b^p$), to the $B_s$. The cost of traversing this path is,

$$cost = dist(R_{Last}, R_s^n) + dist(R_b^p, B_S) \tag{7}$$

**Case 5:** Burst traffic at previous point MS has already visited: The MS will immediately change its trajectory towards the DCP of the burst area and collect the data. After collecting data from the DCP, it will change its directory towards $R_s^n$ and after collecting data from $R_s^n$ it will collect data using regular data collection method for the rest of the DCPs until the round finishes. The cost of traversing this path is ,

$$cost = dist(R_b, R_s^n) \tag{8}$$

After comparing the cases the mobile sink makes decision where to go next and how to finish collecting the sensory data from the efficiently calculated regular DCPs. The MS considers the least amount of path length while making decision in which way it will traverse and finish the round efficiently.

## 3.4 Data Collection

At first the data will be collected in a single-hop fashion from the sensor nodes. The data then will be transmitted to the cluster head (as it's collected via single hop fashion from the nodes to the cluster heads[7],[6]), and after that the collected data will be transferred to the Mobile Sink at the DCPs to minimize the energy consumption. And, in every round the mobile sink transfers the collected data from the DCPs to the $B_s$. The MS transmits the data and collect sensor data from the DCPs in the next rounds, and this process continues.

# 4 Performance Evaluation

## 4.1 Simulation Platform

In this work, the overall simulation is performed using MATLAB R2021b running on a Windows machine with Intel Core i5 $8^{th}$ Gen CPU & 12 GB of RAM. Moreover, all the results were evaluated when the sensor network had the highest number of clusters.
The network deployment parameters are mentioned in table 2.

We used a sensing field of 200m in length and 200 m in width for the network area, for simulation in MATLAB software. We used a varying number of nodes from 50 to 400 for comparing the different parameters mentioned in the figures 8, 9, and 10. The nodes are deployed randomly. The range of the base station is set to 15 meter for the simulation purpose. As the network size is smaller and, most typical communication modules allow for adjustable power settings, limited communication range for both the mobile sink and base station can be set. We used a mobile sink for the network area. As the base sink is electricity-powered the energy is considered to be infinite. The speed of the mobile sink is set to 1.5 m/s which is adjustable in different scenario (for example when busrt traffic occurs).
The energy hole or the hot-spot problem in WSN is solved by using mobile sink and burst-traffic aware routing algorithm. Burst data is collected in an efficient manner on priority basis by the mobile sink, resulting no data loss. Energy consumption is reduced by efficient routing strategies.

| Network Deployment Parameter | Value |
|---|---|
| Sensing field dimensions | 200 x 200 m$^2$ |
| Number of Sensor node ($\mathbb{N}$) | 50-400 |
| Deployment type | Random |
| Number of mobile sinks | 1 |
| Range of the base station ($B_s$) | 15 m |
| Energy of the base station (sink node) | $\infty$ |
| Speed of the Mobile Sink | 1.5 m/s |

**Table 2: Network Deployment Parameters**

| Simulation Parameter | Value |
|---|---|
| Packet size | Bytes/packet |
| Data transmission rate of sensor nodes | 10 pckt/s |
| Simulation rounds | 1400 |
| Energy threshold value ($\alpha$) | 500 J |
| Coverage threshold value ($\beta$) | 10 m to 15 m |
| Coverage range of General Node ($\mathbb{N}_G$) | $\beta$ |
| Initial Energy of General Node ($\mathbb{E}_G$) | $\alpha$ |
| Coverage range of Advance Node ($\mathbb{N}_A$) | 2 $\beta$ |
| Initial Energy of Advance Node ($\mathbb{E}_A$) | 2 $\alpha$ |
| Coverage range of Super Node ($\mathbb{N}_S$) | 3 $\beta$ |
| Initial Energy of Super Node ($\mathbb{E}_S$) | 3 $\alpha$ |

**Table 3: Simulation Parameters**

Table 3 contains the simulation parameters for the sensor nodes deployed in the network. The simulation parameters for the sensor nodes include energy-threshold ($\alpha$) and coverage-threshold ($\beta$) values, which are adjustable based on sensor properties like energy and coverage radius. Typical low-powered sensors have a range of 10 to 100 meters and energy levels of 360 to 1800 joules. In this simulation, the energy threshold is set to 500 J, and the coverage threshold ranges from 10 to 15 meters. These settings allow for an extended network lifetime with multiple routing rounds by the mobile sink (MS) for data collection. The sensor nodes transmit data at a rate of 10 packets per second to the cluster heads, with packet size measured in Bytes/packet.

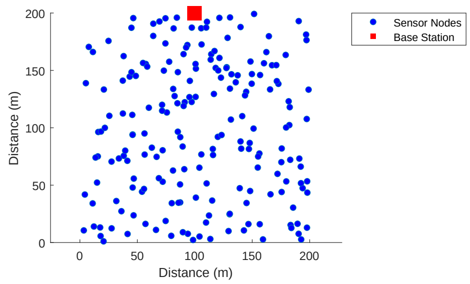


**Figure 4: Sensor Network Deployment**

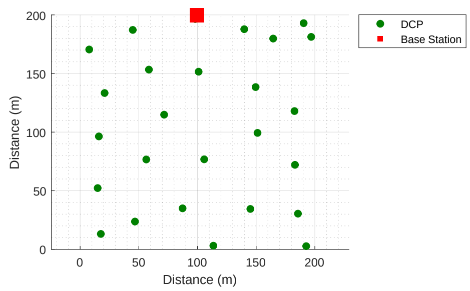


**Figure 5: Data Collection Points (DCPs)**

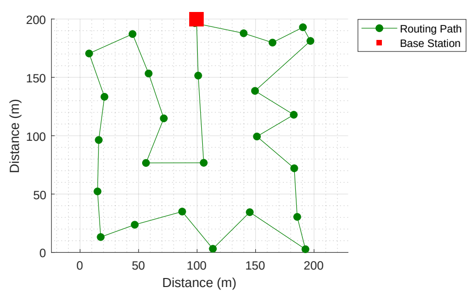


**Figure 6: Routing Path of the Mobile Sink**

**Figure 7: Simulation Environment and Path Routing using MATLAB**

## 4.2 Simulation Results and Analysis

The simulation experiments are conducted with 4 different methods, which are BTAmm[13], HACDC[4], DPPMSBT[15] and our proposed EBTCM method presented in this book. The simulation network's topology is shown in figure 7 where we used 200 sensor nodes in 200x200 sq. meter sensor area. The sub-figure 4 is the sensor network where all the heterogeneous sensor nodes are deployed randomly, the sub-figure 5 shows the selected data collection points calculated by the cluster head eligibility value $\Phi_i$ mentioned in equation 5. The sub-figure 6 shows the regular routing path of the mobile sink from the base station through the DCPs and, finishing a round visiting the base station, and transferring the collected data to it.

*4.2.1 Average Path Length.* The path length refers to the total distance traveled by the mobile sink in a single round, starting and ending at the base station while visiting all data collection points. In Figure 8, the curves for BTAmm [13], DPPMSBT [15], and our proposed EBTCM method are compared. The results show that our EBTCM method achieves the shortest path length, with the slope of its curve being the lowest among all. Specifically, for 200 sensor nodes, the mobile sink travels an average of 1295 meters using our method, compared to 1550 meters in BTAmm and 1950 meters in DPPMSBT. This indicates that our proposed routing strategy is more efficient in minimizing the path length.

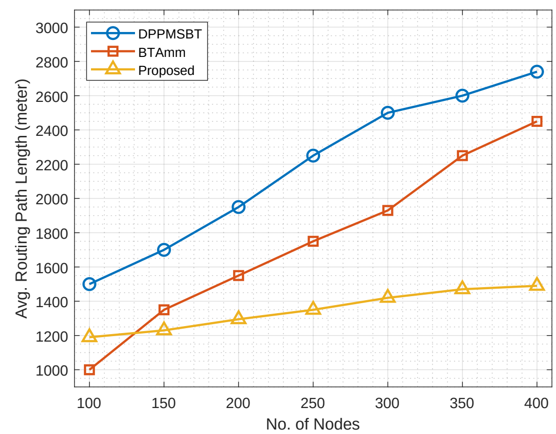


**Figure 8: Average Path Length vs No. of Nodes**

*4.2.2 Network Lifetime.* Network lifetime refers to the lifespan or the survival time of the whole sensor network, and for our work we defined it as from the starting of the simulation till the time of 20% of all of the sensor nodes' death(shortfall of energy of the nodes which causes death to the sensor nodes). In figure 9 we see that network lifetime in sense of number of rounds are increased in comparison with the BTAmm[13], HACDC[4] algorithms. Here, the comparison of data points of the curves of BTAmm[13], HACDC[4], and proposed routing method, our work shows comparatively the highest values of all curves. We see that the overall lifetime of the sensor network (in terms of how many rounds of traversal is completed by the mobile sink) is increased in comparison with BTAmm[13], HACDC[4]. Considering a specific set of data points in the figure 9 an average of 1210 rounds in BTAmm[13] is completed, whereas 1160 rounds in HACDC[4], and only 1250 rounds in our proposed method is completed, in a round against 200 number of sensor nodes, which shows that our proposed method is better considering the lifetime of the sensor network.

*4.2.3 Average Energy Consumption.* From figure 10 we see that average energy consumption of BTAmm[13] and HACDC[4] is comparatively higher than that of our proposed work. The reason behind it is our proposed (EBTCM) method use shorter path than the other two methods, for this, the mobile sink can traverse quickly and collect data in timely manner. When burst data[13] occurs, by using the proposed routing strategy the mobile sink immediately goes to the data collection point and collect the data from the cluster head,

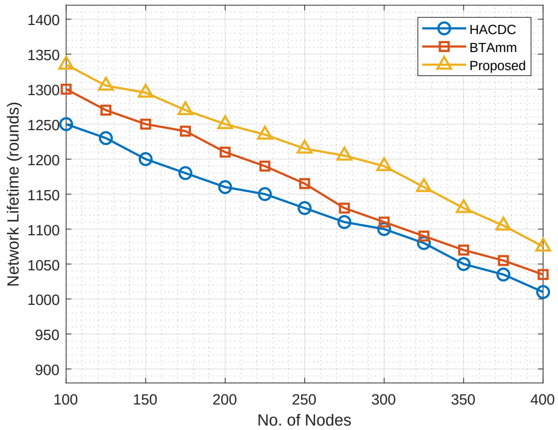


**Figure 9: Network Lifetime**

which reduces the data packet loss in overall system. In figure 10, the comparison of the curves of data points of BTAmm[13], HACDC[4], and proposed routing method, our work shows comparatively lower values in all curves. We see that the average energy consumption's of the sensor network for our proposed method (in terms of energy consumption for data transfer of the sensor nodes and routing of the mobile sink) are comparatively lower in value with respect to BTAmm[13], and HACDC[4].

Considering a set of specific data points in the figure 10 an average of 25.9 Joule of energy is consumed per round in BTAmm[13], whereas 22.9 Joule energy is consumed by HACDC[4], and only 20.90 Joule energy needed per round in our proposed method, in a round against 200 number of sensor nodes, which shows that our proposed method is better considering the average energy consumption of the sensor network.

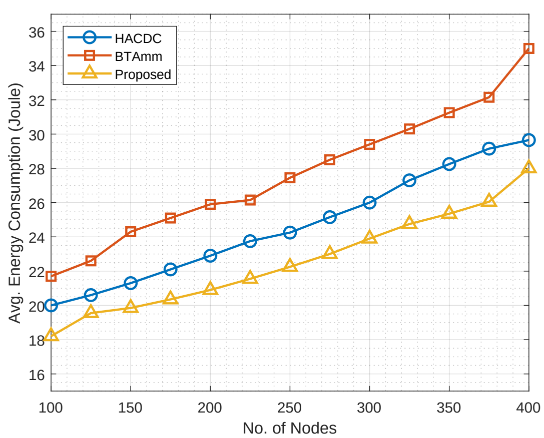


**Figure 10: Average Energy Consumption**

## 5 Conclusion and Future Work

In this work, we have presented a comprehensive overview of existing research on energy-efficient data collection, routing, and sink mobility management in WSNs. Through a thorough review of the literature, we identified several challenges and shortcomings in current approaches, including energy inefficiency, packet loss, and latency issues.

To address these challenges, we proposed novel algorithms and techniques aimed at improving the scalability, reliability, and energy efficiency of WSNs. Our contributions include the development of an energy-efficient data collection approach, an efficient routing strategy for multiple mobile sinks, and innovative clustering and sink mobility management techniques.

Overall, our research contributes to advancing the state-of-the-art in WSNs and lays the foundation for more efficient and reliable data collection, routing, and sink mobility management in these networks.

In future if more durable power source is ensured, more complex calculations can be accomplished to make choices while forming the clusters and cluster heads, and select the route.

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